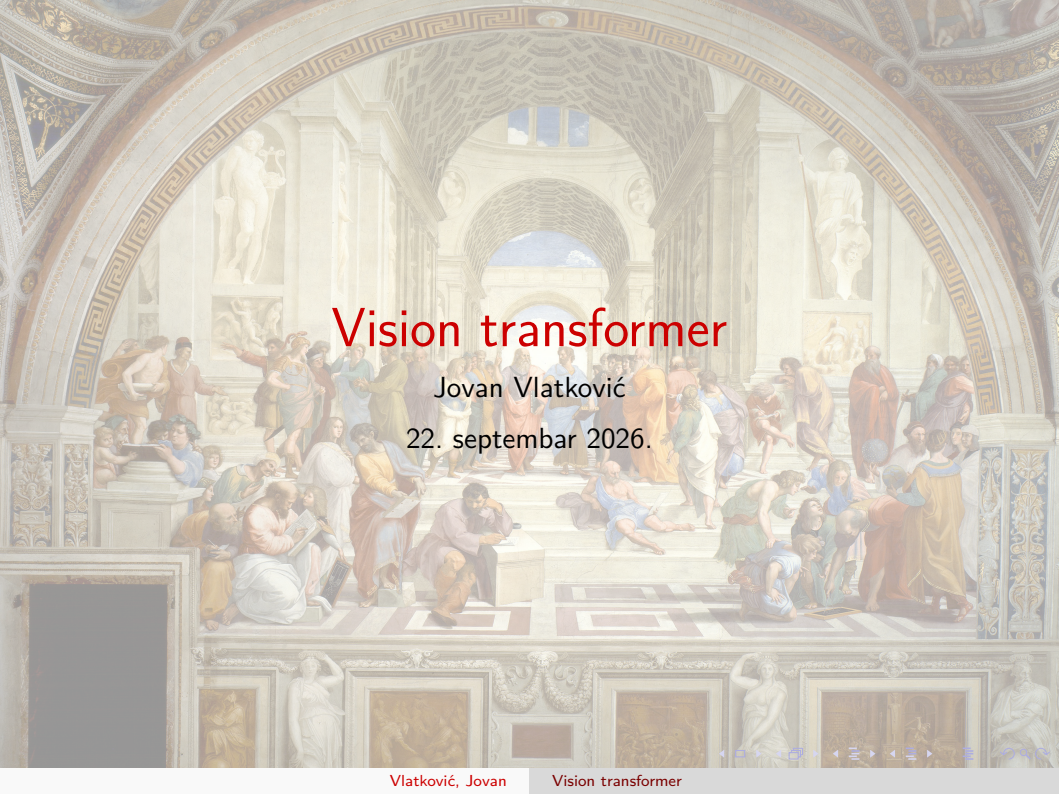


The background of the slide is a detailed reproduction of the fresco 'The School of Athens' by Raphael. It depicts a group of ancient Greek philosophers in a grand, vaulted hall. In the center, Plato and Aristotle are the focal points. Plato is shown as an older man with a long white beard, wearing a blue robe, pointing his right index finger towards the sky. Aristotle is a younger man with a dark beard, wearing a red robe, gesturing with his right hand palm-down towards the earth. They are surrounded by other philosophers in various poses, some seated and some standing, engaged in discussion or study. The architecture features high arches, statues in niches, and a checkered floor.

Vision transformer

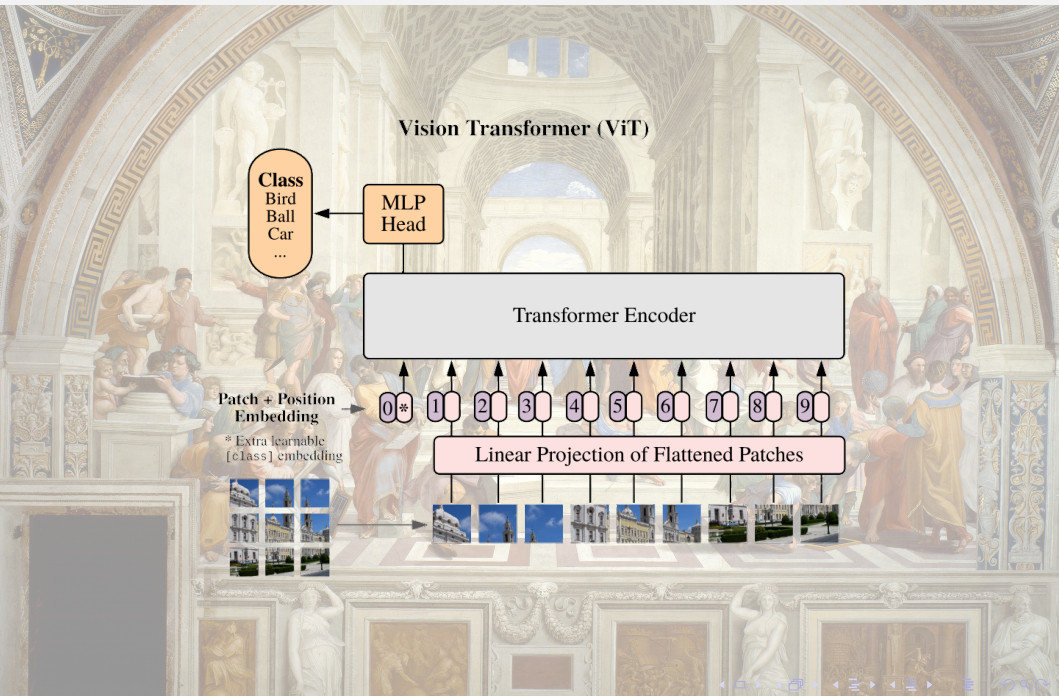
Jovan Vlatković

22. septembar 2026.

Vision transformer (ViT)

- ▶ Adaptacija arhitekture transformera za primenu u obradi i analizi slika
- ▶ Attention Is All You Need
- ▶ An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale
- ▶ Koristi se za klasifikaciju slika, detekciju objekata, ...

Arhitektura ViT



Arhitektura ViT - podela slike na delove (patches) i vektorizacija

- ▶ Slika je oblika $x \in \mathbb{R}^{H \times W \times C}$
- ▶ Deli se u N delova $x_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$
- ▶ Enkoder koristi vektore dimenzije D, pa se svako x_p vektorizuje naučivom linearnom projekcijom $z_0^i = x_p^i E, \forall i \in \{1, \dots, N\}$
- ▶ $E \in \mathbb{R}^{(P^2 \cdot C) \times D}$

Arhitektura ViT - poziciono enkodiranje i CLS token

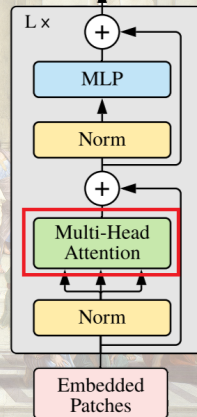
- ▶ Pozicionim enkodiranjem se čuvaju prostorne informacije značajne zbog deljenog vizuelnog konteksta između susednih delova
- ▶ Koriste se naučivi vektori koji su kolone matrice $E_{pos} \in \mathbb{R}^{(N+1) \times D}$
- ▶ Klasifikacioni (CLS) token ($z_0^0 = x_{CLS}$)
- ▶ Zahvaljujući mehanizmu samopažnje CLS token uzima u obzir informacije sadržane u ostalim tokenima i time formira celokupnu reprezentaciju slike

Arhitektura ViT - enkoder blok

Arhitektura enkodera

- ▶ $\mathbf{z}_0 = [\mathbf{x}_{CLS}; \mathbf{x}_p^1 E; \dots; \mathbf{x}_p^N E] + E_{pos}$
- ▶ $\mathbf{z}'_l = \text{MSA}(\text{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1}, l = 1 \dots L$
- ▶ $\mathbf{z}_l = \text{MLP}(\text{LN}(\mathbf{z}'_l)) + \mathbf{z}'_l, l = 1 \dots L$

Transformer Encoder



Arhitektura ViT - enkoder blok

Normalizacioni sloj

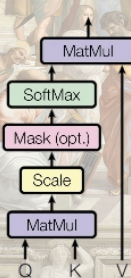
- ▶ Za token $\mathbf{z}_l^i = (a_{l,1}^i, \dots, a_{l,D}^i)$ računamo srednju vrednost i standardnu devijaciju
- ▶ $\mu_l^i = \frac{1}{D} \sum_{j=1}^D a_{l,j}^i$ $\sigma_l^i = \sqrt{\frac{1}{D} \sum_{j=1}^D (a_{l,j}^i - \mu_l^i)^2}$
- ▶ $LN(\mathbf{z}_l^i; \alpha_l^i, \beta_l^i) = \frac{(\mathbf{z}_l^i - \mu_l^i)}{\sigma_l^i} \odot \alpha_l^i + \beta_l^i$

Arhitektura ViT - enkoder blok

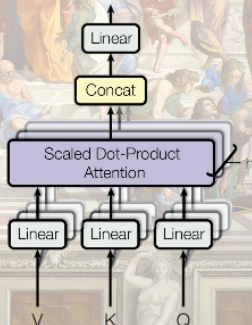
Mehanizam pažnje

- ▶ Omogućava svakom tokenu u sekvenci da odredi koji od ostalih tokena sadrže relevantne informacije za razumevanje konteksta i formiranje njegove nove reprezentacije
- ▶ $Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V$

Scaled Dot-Product Attention



Multi-Head Attention

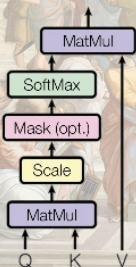


Arhitektura ViT - enkoder blok

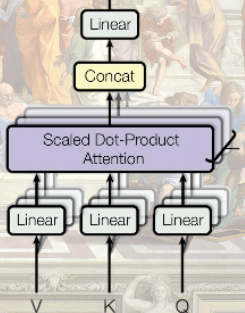
Multi-head self-attention sloj

- ▶ $head_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$
- ▶ $\text{MultiHead}(Q, K, V) = \text{Concat}(head_1, \dots, head_h)W^O$

Scaled Dot-Product Attention



Multi-Head Attention



Arhitektura ViT - enkoder blok

Višeslojni perceptron (MLP)

- ▶ Potpuno povezana neuronska mreža
- ▶ $MLP(\mathbf{x}) = W_2 \text{ GELU}(W_1 \mathbf{x} + b_1) + b_2$
- ▶ $\text{GELU}(x) \approx 0.5 \cdot x \cdot \left(1 + \tanh\left(\sqrt{\frac{2}{\pi}} \cdot (x + 0.044715 \cdot x^3)\right)\right)$

Arhitektura ViT - klasifikaciona glava

- ▶ CLS token sadrži reprezentaciju slike $\mathbf{y} = LN(\mathbf{z}_L^0)$
- ▶ Koristi se neuronska mreža na čijem su izlazu: $\text{logits} = W_{cls} \mathbf{y} + b_{cls}$

Eksperiment

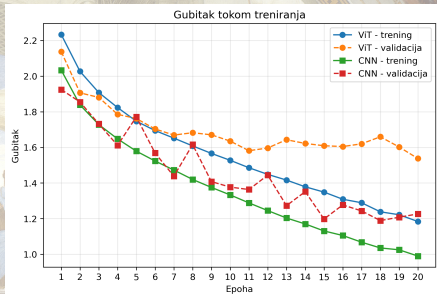
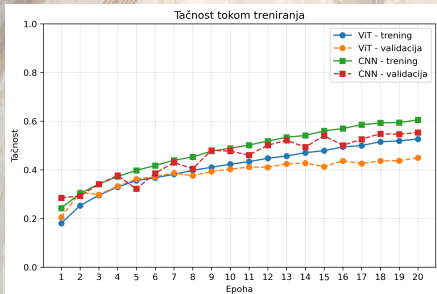
Klasifikacija nad skupom podataka:

- ▶ WikiArt Art Movements/Styles
- ▶ 42500 slika rezolucije 224×224
- ▶ 13 klasa: 'Academic_Art', 'Art_Nouveau', 'Baroque', 'Expressionism', 'Japanese_Art', 'Neoclassicism', 'Primitivism', 'Realism', 'Renaissance', 'Rococo', 'Romanticism', 'Symbolism', 'Western_Medieval'
- ▶ Delimo na trening i test skup smanjenog obima

Eksperiment - postavka

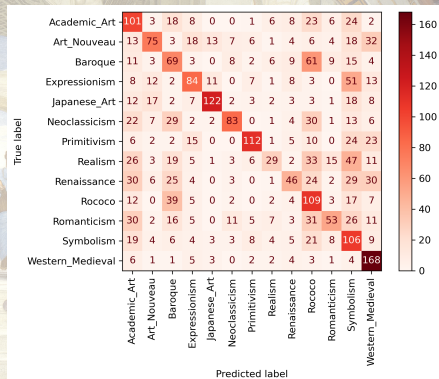
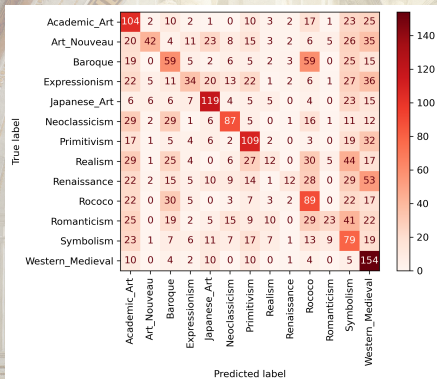
- ▶ Treniramo ViT i CNN model
- ▶ Adam optimizator i Cross-entropy loss funkciju gubitka
- ▶ ViT: $L = 4$, $h = 4$, $D = 128$, $N = 196$
- ▶ CNN: pet uzastopnih konvolutivnih blokova - konvolutivni sloj sa jezgrom dimenzije 3×3 , normalizacioni sloj (batch normalization), ReLU aktivaciona funkcija i sloj Max pooling dimenzije 2×2
- ▶ 20 epoha sa jednostavnom augmentacijom

Eksperiment - rezultati



Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
ViT	0.355	0.401	0.355	0.326	0.326
CNN	0.445	0.494	0.445	0.441	0.441

Eksperiment - rezultati



Eksperiment - diskusija

- ▶ CNN model nadmašuje ViT u relevantnim metrikama
- ▶ Veličina modela i skupa podataka
- ▶ Preprilagođavanje modela
- ▶ Augmentacija

The background of the slide is Raphael's famous fresco 'The School of Athens'. It depicts a group of ancient Greek philosophers in a grand, vaulted hall. Plato and Aristotle are at the center, surrounded by other figures like Pythagoras, Euclid, and Socrates. The architecture features high arches, statues in niches, and a checkered floor. The text 'Hvala na pažnji!' is superimposed in the center in a red, sans-serif font.

Hvala na pažnji!